

Smart Tourism System

Ho Chi Minh City - Computational Thinking Application

Group 2 - Class 24C11

Supervisor: Dr. Bui Tien Len

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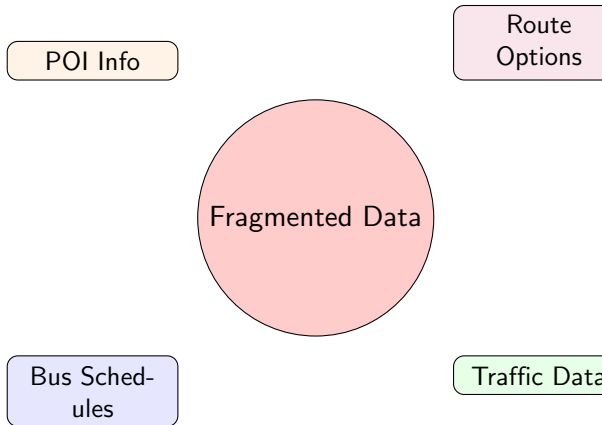
Outline

- 1 Problem Analysis
- 2 Computational Thinking
- 3 Technical Implementation
- 4 Testing & Results
- 5 Conclusion

The Problem: Urban Tourism Challenges

Travelers face significant friction:

- ❌ Too many choices, not enough clarity
- ❌ Time lost to planning instead of exploring
- ❌ Stress in unfamiliar environments
- ❌ Late reactions to delays & congestion
- ❌ Fewer experiences, lower satisfaction



Project Goals

Our Solution: AI-Powered Tourism Platform

A comprehensive system integrating multiple data sources into one unified interface.



AI Trip Planning

Natural language
itinerary generation



Bus Routing

A* pathfinding with
transfers



Live Congestion

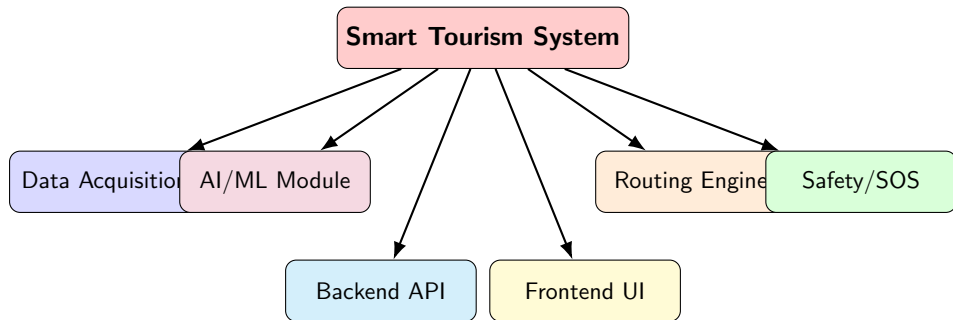
YOLO detection from
700+ cameras



SOS System

Community safety
alerts

Decomposition: System Modules



Pattern Recognition

Key patterns identified across modules:

Geographic Proximity

- Bus stop $\leftrightarrow$ User location
- Camera $\leftrightarrow$ Road segment
- POI $\leftrightarrow$ User query

Solution: Haversine distance function

Caching for Performance

- Graph construction: 30s $\rightarrow$ <1s
- Fuzzy match index
- Pickle serialization

Threshold-Based Classification

- Vehicle count $\rightarrow$ Congestion level
- P50, P80 percentiles

Graceful Degradation

- Modal GPU fails $\rightarrow$ Local CPU
- TomTom fails $\rightarrow$ Cached route

Abstraction: Unified Interfaces

Place Abstraction

- name, coordinates, category
- normalized_name (for matching)
- Works for: restaurants, hotels, landmarks

Graph Abstraction

- Nodes = Bus stops
- Edges = Routes between stops
- Weight = Time + transfer penalty

Detection Result Abstraction

- camera_id, timestamp
- counts: {car, motorcycle, bus, truck}
- processing_time, method

API Response Abstraction

- Consistent JSON structure
- Status codes: 200, 400, 404, 500
- Error messages included

Algorithm Design: A* Bus Routing

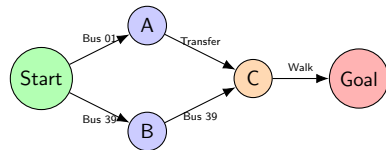
Modified A* Algorithm:

- 1 Find nearby stops (walk $\leq 600\text{m}$)
- 2 Build priority queue with $f = g + h$
- 3 Expand nodes by bus routes
- 4 Apply transfer penalty (+5 min)
- 5 Reconstruct path when goal reached

Cost Function:

$$f(n) = g(n) + h(n)$$

where $h(n)$ = Haversine distance to goal



Data Sources

Source	Type	Data
HCMC Traffic Portal	Live API	702 traffic cameras
TomTom API	Commercial	Geocoding, routing
HCMC Bus Map	Enterprise	Routes, stops, schedules
Groq API	LLM Service	Trip planning queries
Custom Dataset	Manual	700+ POIs

Storage Strategy:

- MySQL: User data (ACID compliance)
- SQLite: Detection results (fast writes)
- JSON: Camera locations, POI database
- CSV: Bus network (pandas compatible)

YOLOv11 Vehicle Detection

- Input: 960×540 camera images
- Classes: car, motorcycle, bus, truck
- CLAHE preprocessing
- Modal GPU deployment (T4)

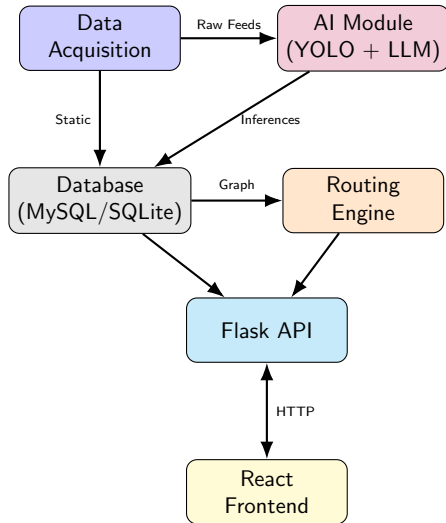
Groq LLM (Llama 3.3 70B)

- Natural language trip queries
- Structured JSON output
- 1-2s response time
- HCMC context in prompt

RapidFuzz Matching

Bridges LLM output to database: "Ben Thanh" → Ben Thanh Market (ID: 42)

System Architecture



Testing Results

Authentication

- Register/Login: ✓ Pass
- OAuth (Google/GitHub): ✓ Pass
- Invalid input handling: ✓ Pass

Vehicle Detection

- Modal GPU: ✓ Working
- Local fallback: ✓ Working
- Accuracy: Good on clear images

Bus Routing

- Valid routes: ✓ Pass
- Edge cases: ✓ Handled
- Response: 7000 VND, 21-36 min

AI Trip Planner

- Query parsing: ✓ Pass
- Fuzzy matching: ✓ 85%+ match
- Graceful degradation: ✓ Pass

Performance Metrics

Metric	Before	After Optimization
Graph Load Time	30s	<1s
YOLO Inference (CPU)	5s/image	–
YOLO Inference (GPU)	–	0.5s/image
LLM Response	–	1-2s
Bus Route Calculation	–	<2s

Key Optimizations:

- Pickle caching for graph data
- Modal serverless GPU deployment
- Pre-computed fuzzy match indices

Project Value

- ① **Reduces commute stress** – Optimal bus routes with accurate estimates
- ② **Improves traffic awareness** – Real-time congestion from 100+ cameras
- ③ **Enhances tourism** – AI trip planning in natural language
- ④ **Enables community safety** – SOS alert system

CT Lessons Learned

- **Decomposition** enabled parallel development
- **Pattern Recognition** led to reusable functions
- **Abstraction** made the system scalable
- **Algorithm Design** required careful trade-offs

Team Members

ID	Name	Role
24127021	Nguyễn Thành Đạt	Routing & SOS Mapping
24127041	Hoàng Trung Hiếu	Bus Mapping & Recommendation
24127298	Lương Hưng Phát	Traffic Congestion Analysis
24127188	Nguyễn Minh Khoa	Data Acquisition
24127430	Nguyễn Anh Khôi	Database Engineering
24127373	Nguyễn Tấn Hiệu	UI/UX & Integration

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Thank You!

Questions?

 GitHub: [Project Repository]

 Demo: [Live Demo URL]